

Indian Statistical Institute
B. Statistical Data Science 2nd Year
Linear Statistical Models
Class Test - 1
27 February 2026

Name: _____

Total Marks: 10

Roll Number: _____

Instructions

- Answer as much as you can. The maximum you can score is 10.
- Only write the answer in the space provided or tick the correct option. Any additional writing will not be graded.
- Rough works should be done on spare pages.

Question 1. Which of the following are linear statistical models?

[4 × 0.5 marks]

- (A) $\mathbb{E}(y) = \alpha + \beta x + \gamma e^{5x^2}$
- (B) $\mathbb{E}(y) = \beta_{11}x_1^2 + \beta_{12}x_1x_2 + \beta_{22}x_2^2$
- (C) $\mathbb{E}(y) = (\beta_1x_1 + \beta_2x_2)^2$
- (D) $\mathbb{E}(y) = \begin{cases} \alpha & \text{if } x < 50, \\ \alpha + \beta & \text{if } 50 \leq x < 75, \\ \beta + \gamma & \text{if } x \geq 75 \end{cases}$

Question 2. Consider the linear regression model $y = X\theta + \epsilon$ where X is an $n \times d$ matrix and $\theta \in \mathbb{R}^d$. Which of the following conditions ensures that the normal equations $X^\top X\theta = X^\top y$ have a unique solution?

[0.5 marks]

- (A) $\text{rank}(X) = \text{rank}(X|y)$
- (B) $\text{rank}(X) = n$
- (C) $\text{rank}(X) = d$
- (D) $\text{rank}(X^\top X) = n$

Question 3. Let P be an orthogonal projection matrix onto a proper subspace of $\mathbb{R}^n$. Which of the following is/are FALSE?

[1 mark]

- (A) $(I - P)$ is symmetric and idempotent.
- (B) $(I - P)$ is positive semi-definite.
- (C) P and $(I - P)$ are orthogonal to each other.
- (D) $(I - P)$ is positive definite.

Question 4. Let P be an $n \times n$ orthogonal projection matrix. Identify the correct statement(s) from the following.

[1 mark]

- (A) $P^2v = v \quad \forall v \in \mathbb{R}^n$
- (B) $P^2v = Pv \quad \forall v \in \mathbb{R}^n$
- (C) $P^3v = v \quad \forall v \in \mathbb{R}^n$
- (D) $P^3v = Pv \quad \forall v \in \mathbb{R}^n$

Question 5. Let X be an $n \times d$ matrix with $\text{rank}(X) = d$. Which of the following is the orthogonal projection matrix onto $\mathcal{C}(X)$? [0.5 marks]

- (A) $XX^\top$
- (B) $(X^\top X)^{-1}$
- (C) $X(X^\top X)^{-1}X^\top$
- (D) $I - X(X^\top X)^{-1}X^\top$

Question 6. Based on the data $(x_1, y_1), \dots, (x_n, y_n)$ two practitioners are interested in fitting the model $y = \alpha + \beta x + \epsilon$. The first practitioner estimates α and β by the least-squares method. The second practitioner also uses the method of least-squares, but after mean-centering x and y . Find the correct statement about the estimates obtained by the two practitioners.

- (A) Estimates of both the parameters are the same.
- (B) The estimated intercepts are the same, but the estimated slopes may be different.
- (C) The estimated slopes are the same, but the estimated intercepts may be different.
- (D) Estimates of both the parameters are different.

Question 7. Consider the simple linear regression model $y_i = \alpha + \beta x_i + \epsilon_i$, $i = 1, \dots, n$. The fitted least-squares regression line based on 8 pairs of observations $(x_1, y_1), \dots, (x_8, y_8)$ is $\hat{y} = 2 + 0.5x$. Also, $\sum_{i=1}^8 (x_i - \bar{x})^2 = 8$ and $\sum_{i=1}^8 (y_i - \bar{y})^2 = 4$. Which of the following is the value of the coefficient of determination R^2 ? [1.5 marks]

- (A)
- (B)
- (C) 0.50
- (D) 0.75

Question 8. Let $v_1 = (1, 2, -1)^\top$ and $v_2 = (-1, 1, 0)^\top$. Determine scalars a_1 and a_2 such that $a_1v_1 + a_2v_2$ is the orthogonal projection of $(1, 2, 3)^\top$ onto the span of v_1 and v_2 . [1.5 marks]

Question 9. Consider a linear regression setup: $y_i = \alpha + \beta x_i + \epsilon_i$, $i = 1, \dots, 40$, with $\epsilon_i \sim i.i.d.$ We also know that $\sum_{i=1}^{40} x_i^2 = 60$ and $\sum_{i=1}^{40} x_i = 20$. Let $(\hat{\alpha}, \hat{\beta})$ be the OLS estimator of (α, β) . Determine $\text{Var}(\hat{\alpha})$ and $\text{Var}(\hat{\beta})$. [1.5 + 1.5 marks]